PEARL Power and Energy Analysis and Research Laboratory

Escuela de Ingeniería Industrial y Aeroespacial Toledo
Universidad de Castilla-La Mancha

UCLM Universidad Castilla-La Mancha

Tools for the Optimal Operation and Planning of Power Systems

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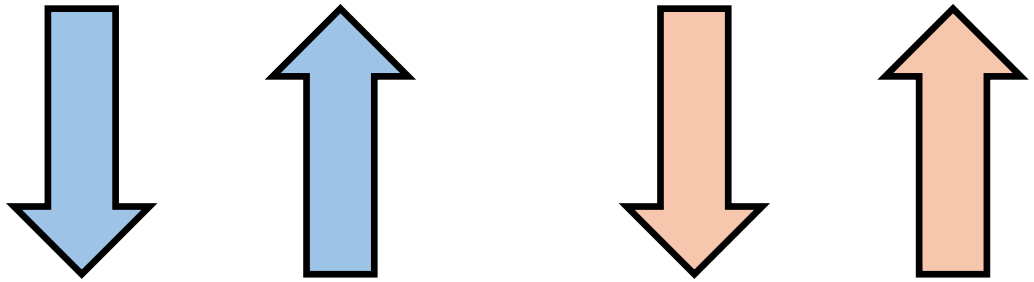
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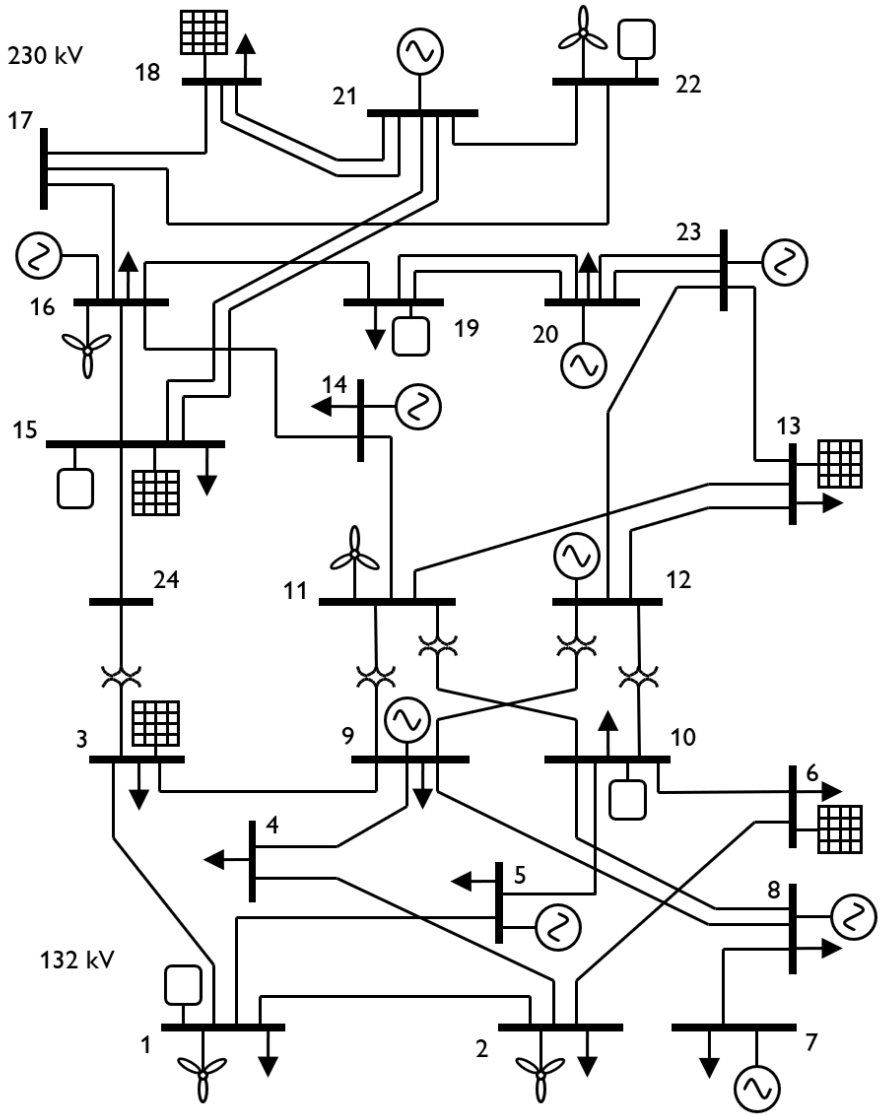
What is this course about?

Address the operation and planning of modern power systems using optimization tools

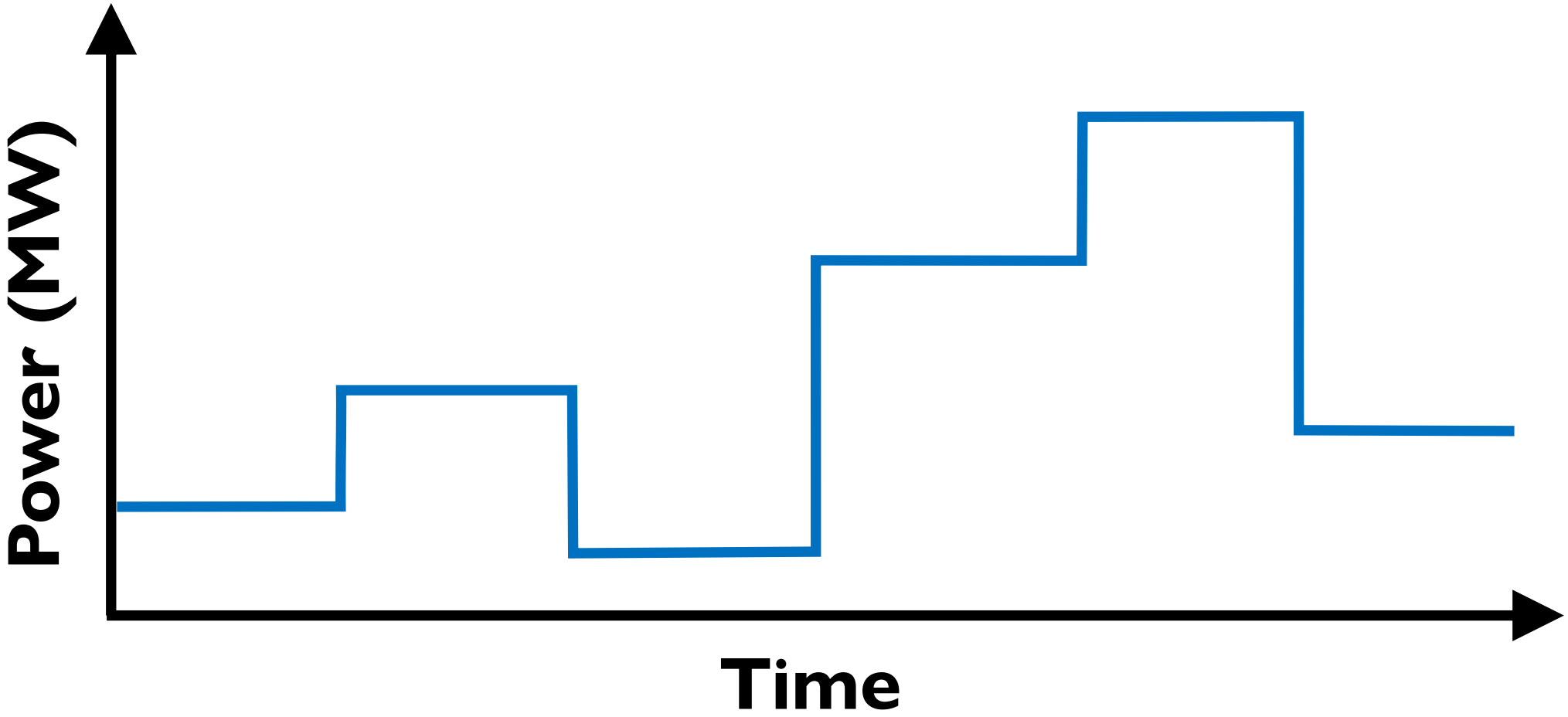
Balance of energy

Generation = Consumption

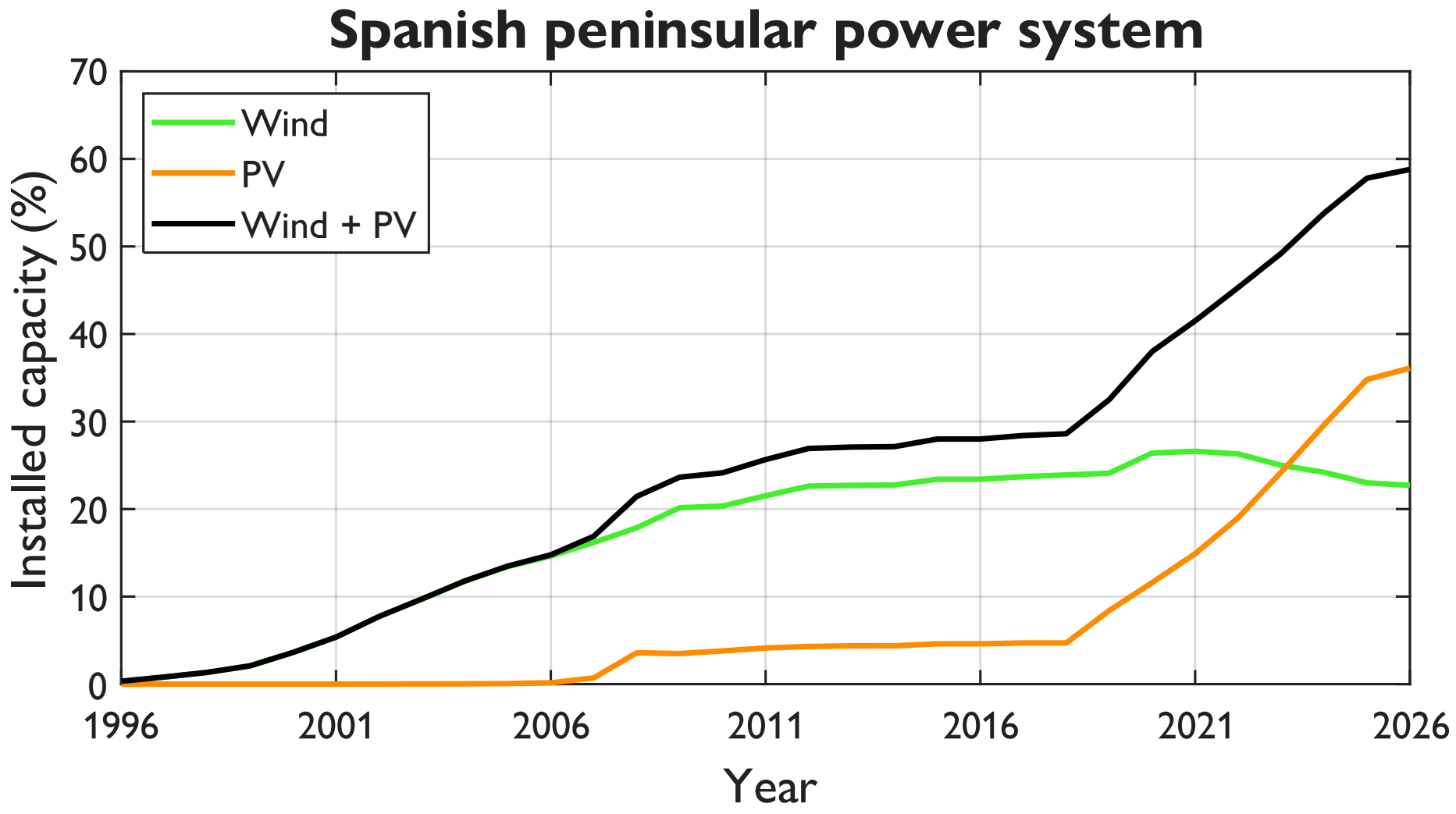




Scheduled production

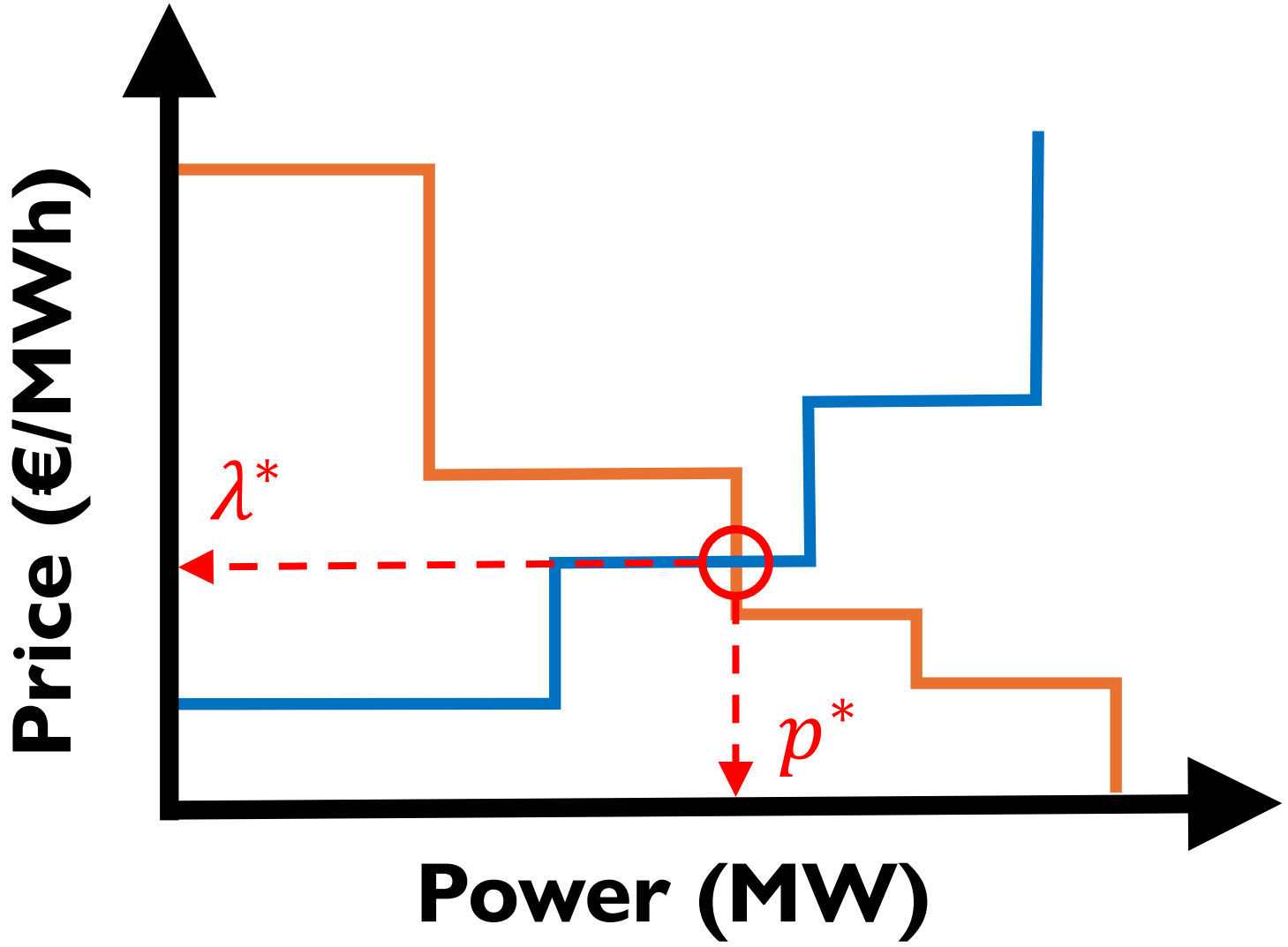


Modern power systems

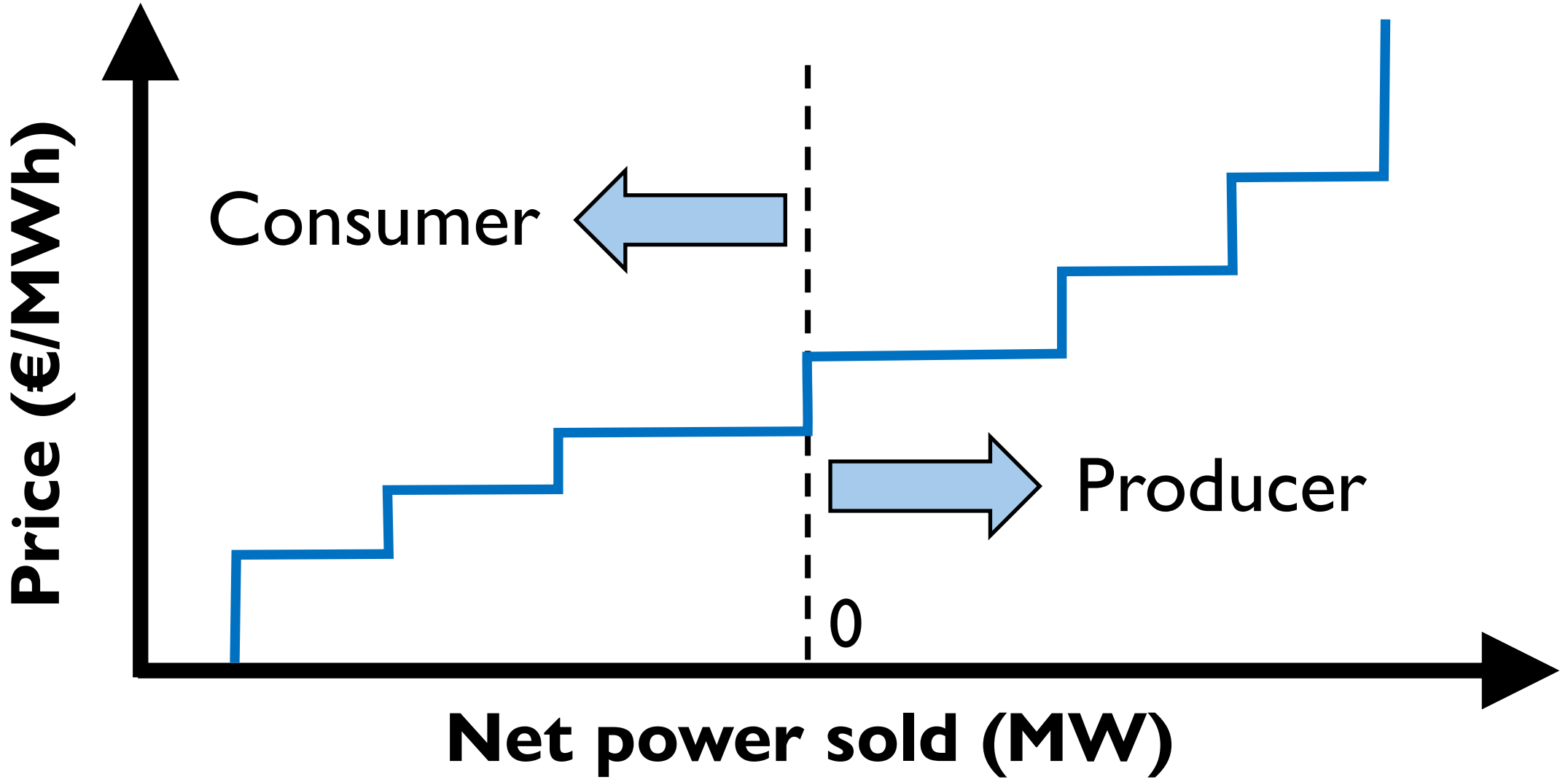


Source: https://www.ree.es/en/datos/generation/installed-capacity-generation

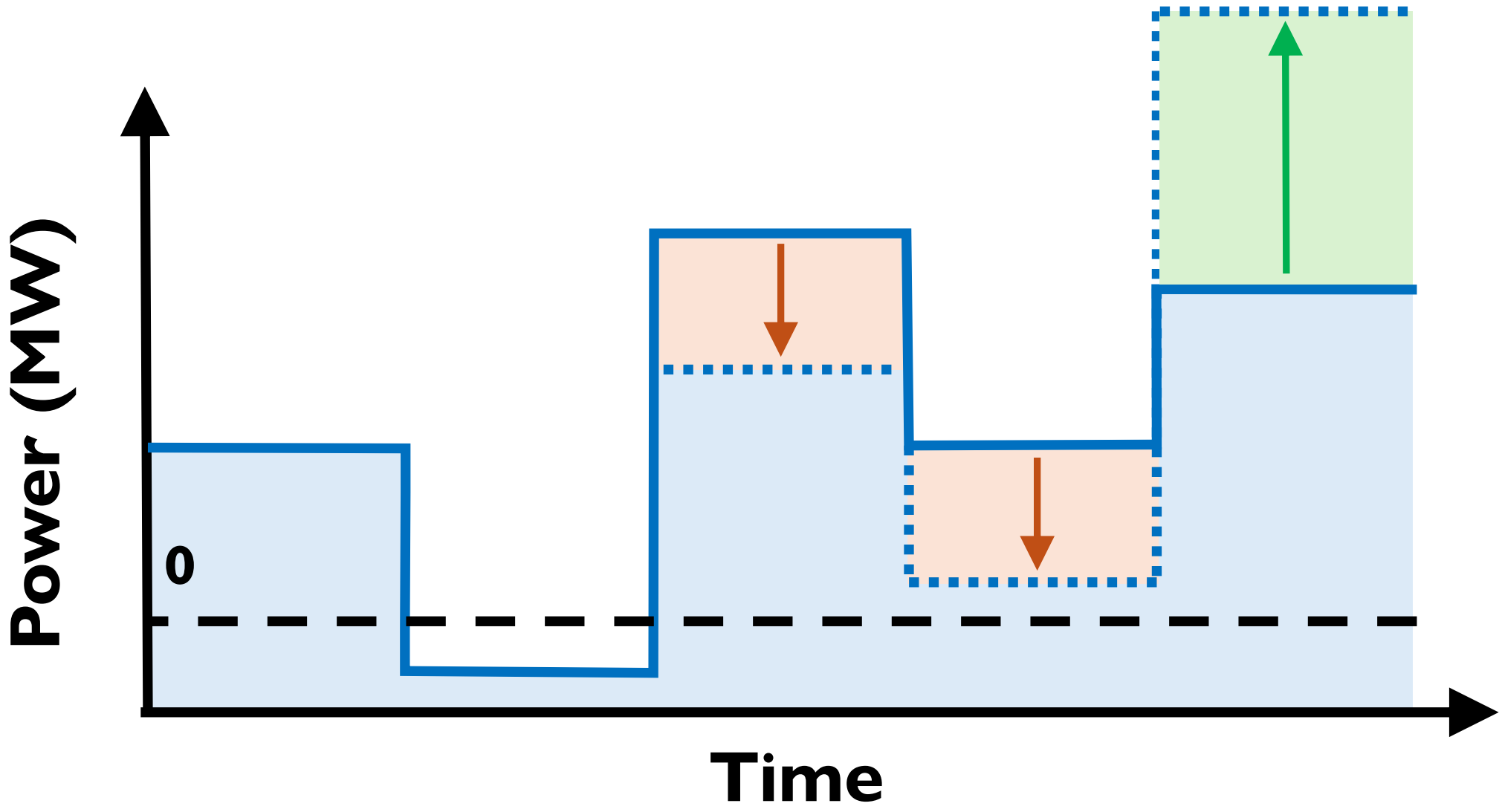
Day-ahead electricity market



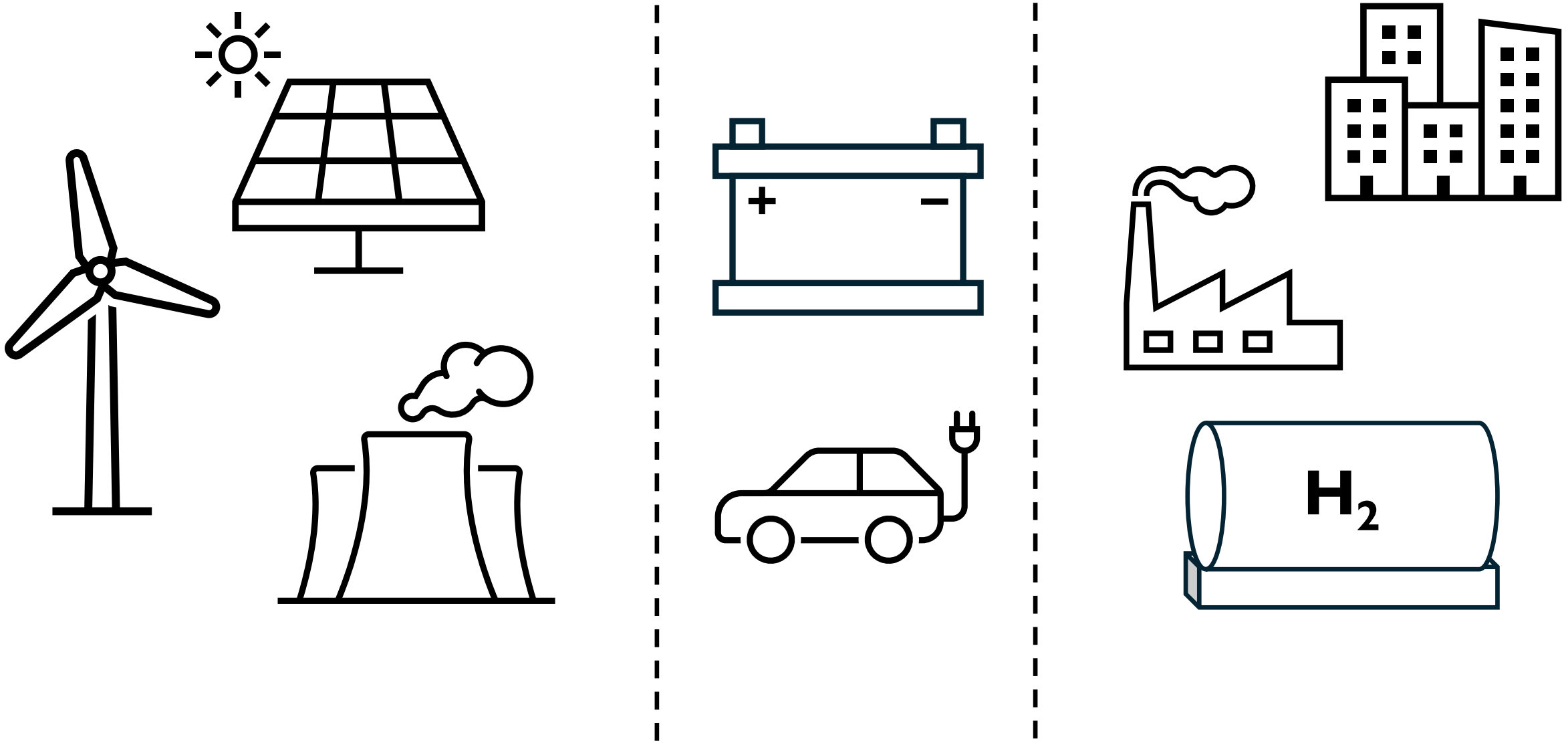
Bidding and offering curves



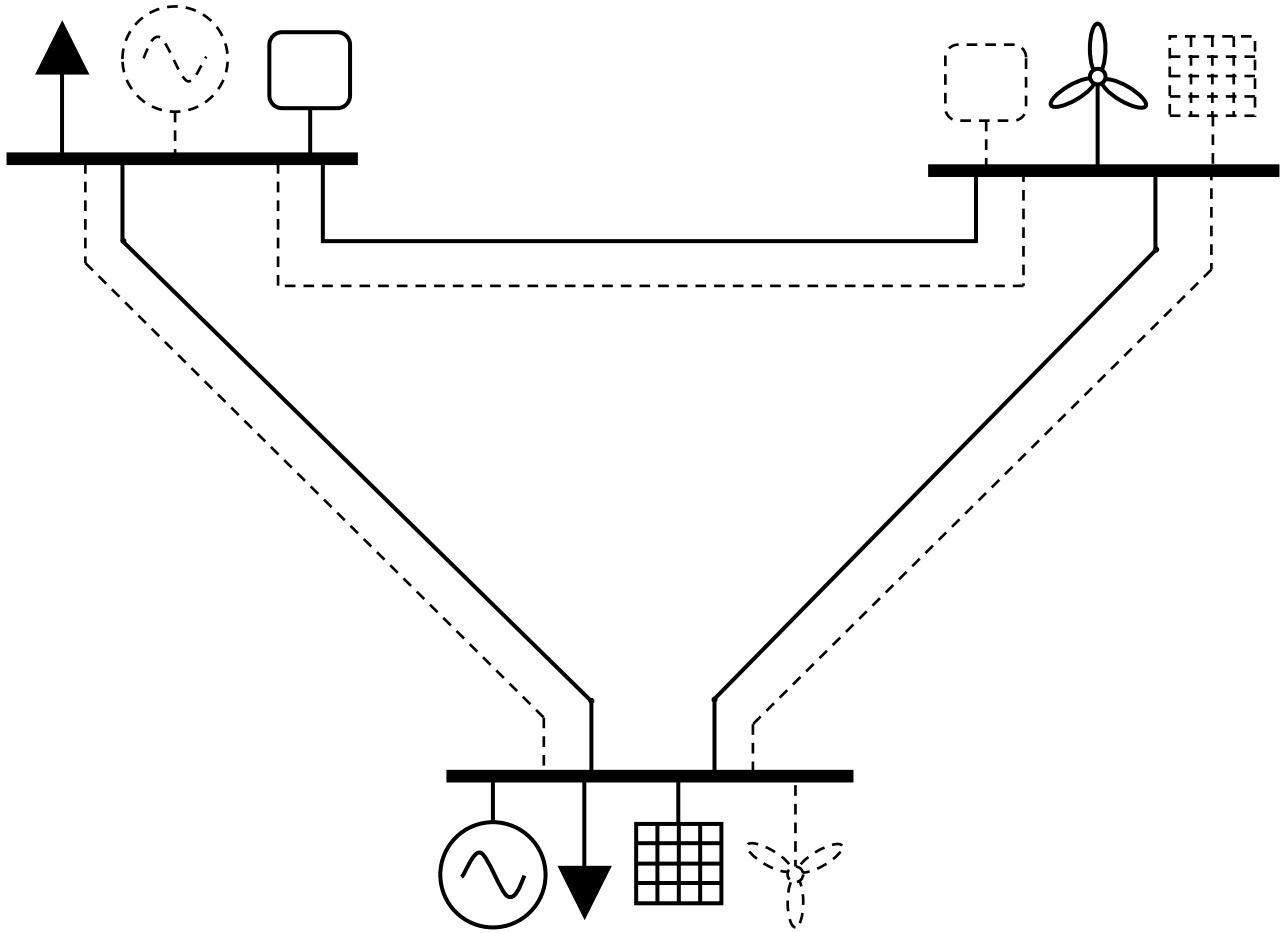
Demand response programs



Virtual Power Plant

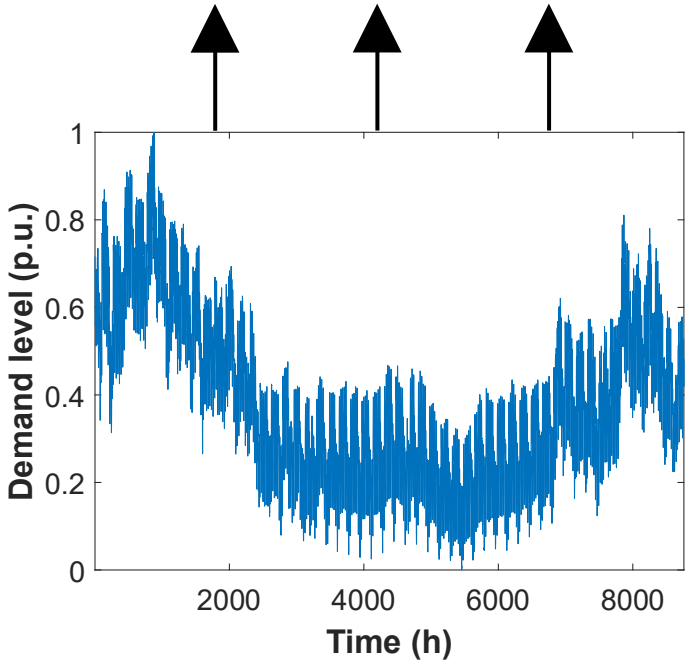


Generation and transmission network expansion

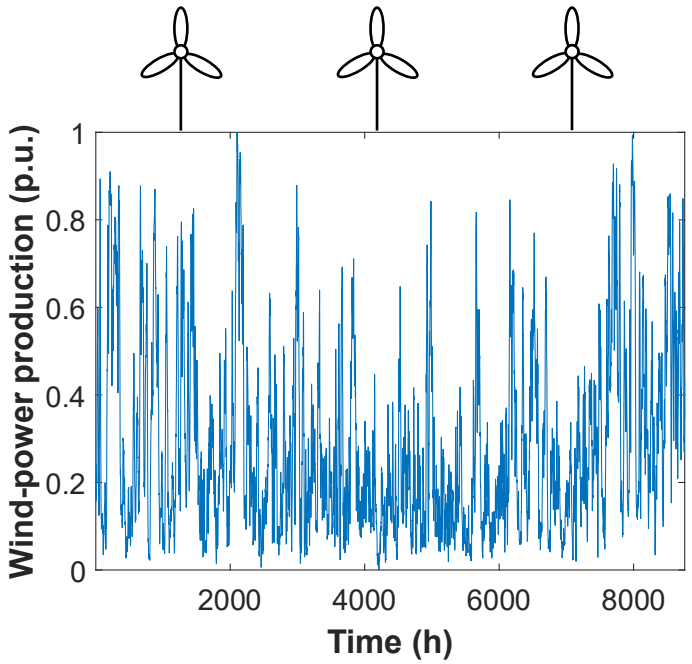


Operational variability

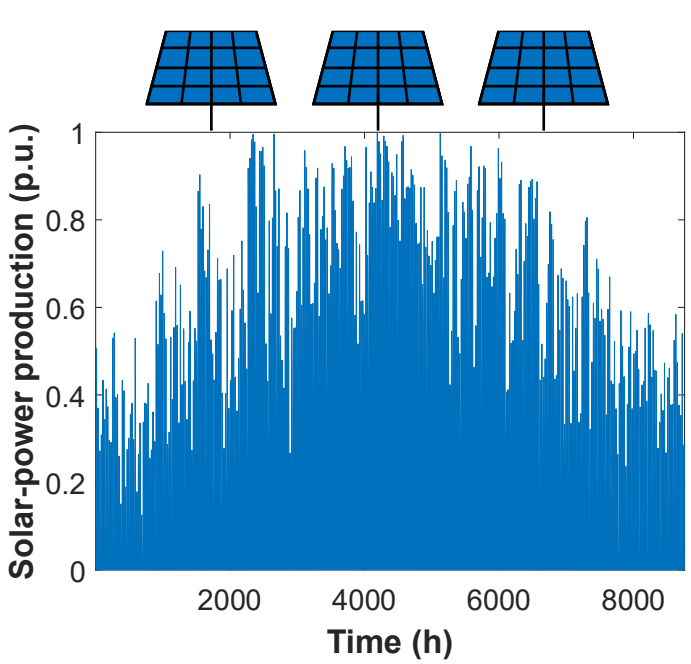
Electrical demand



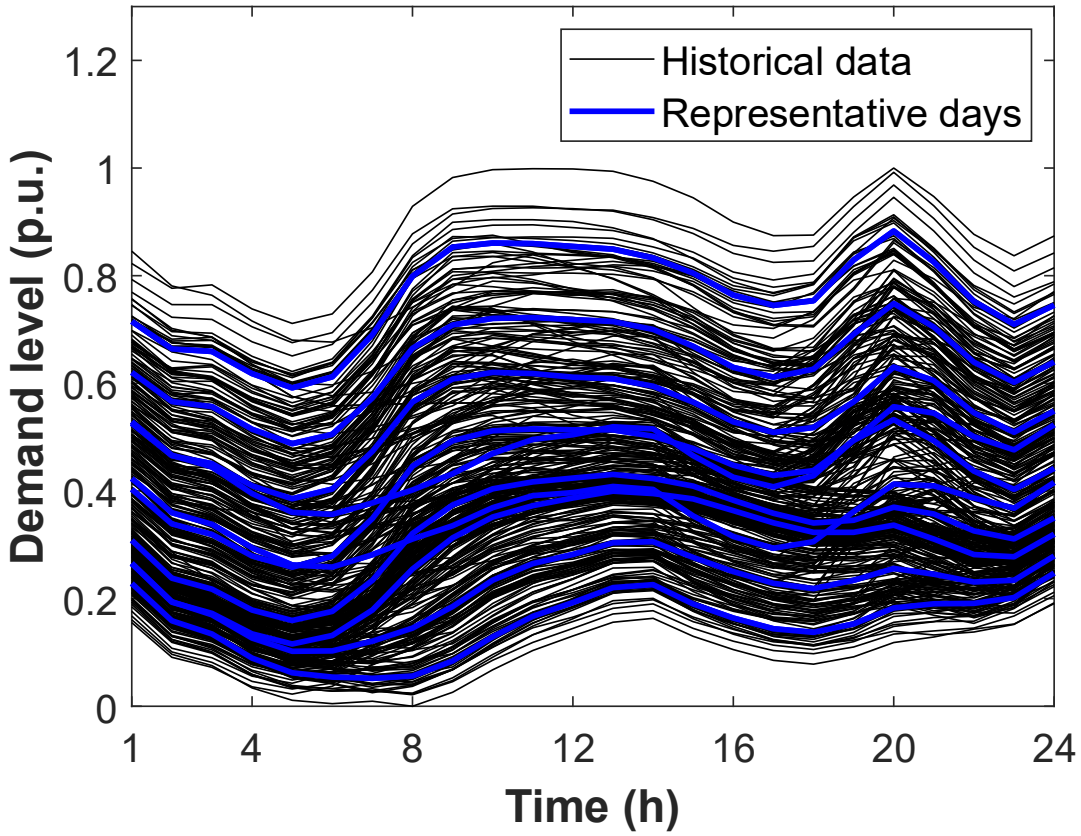
Wind-power production

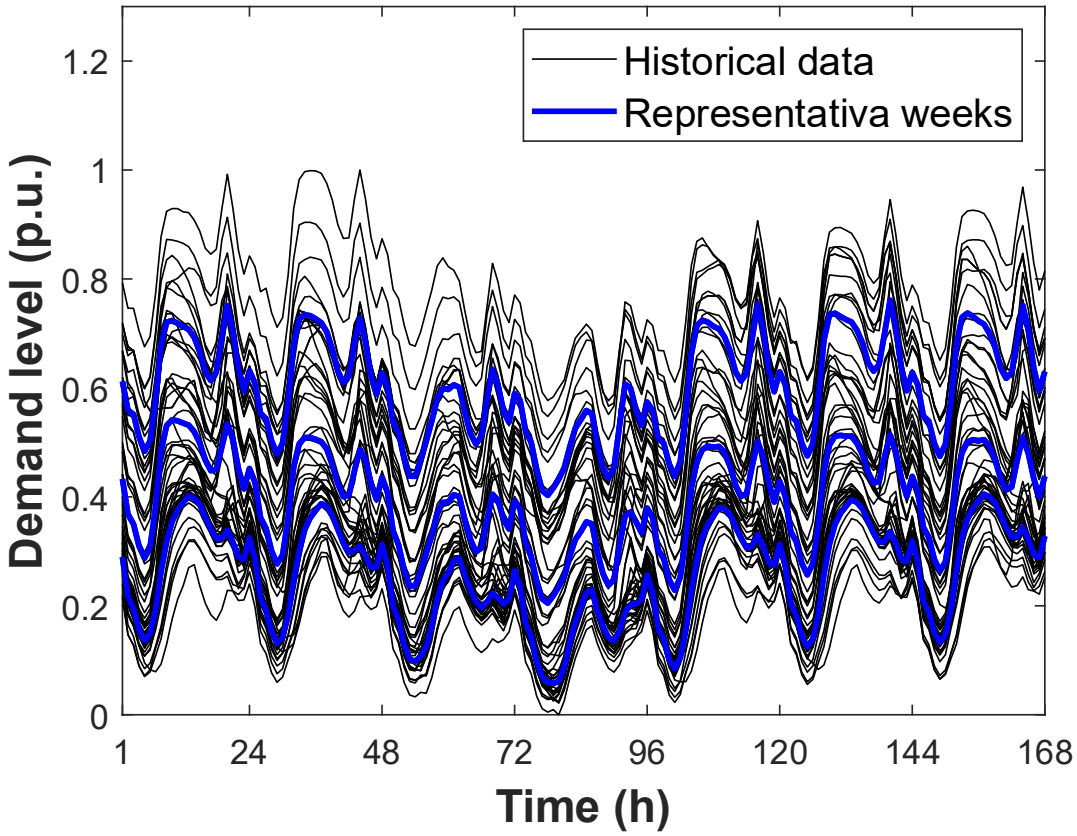


Solar-power production

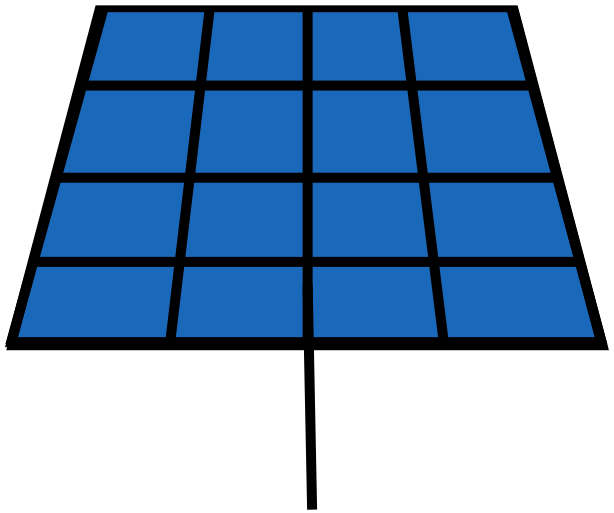


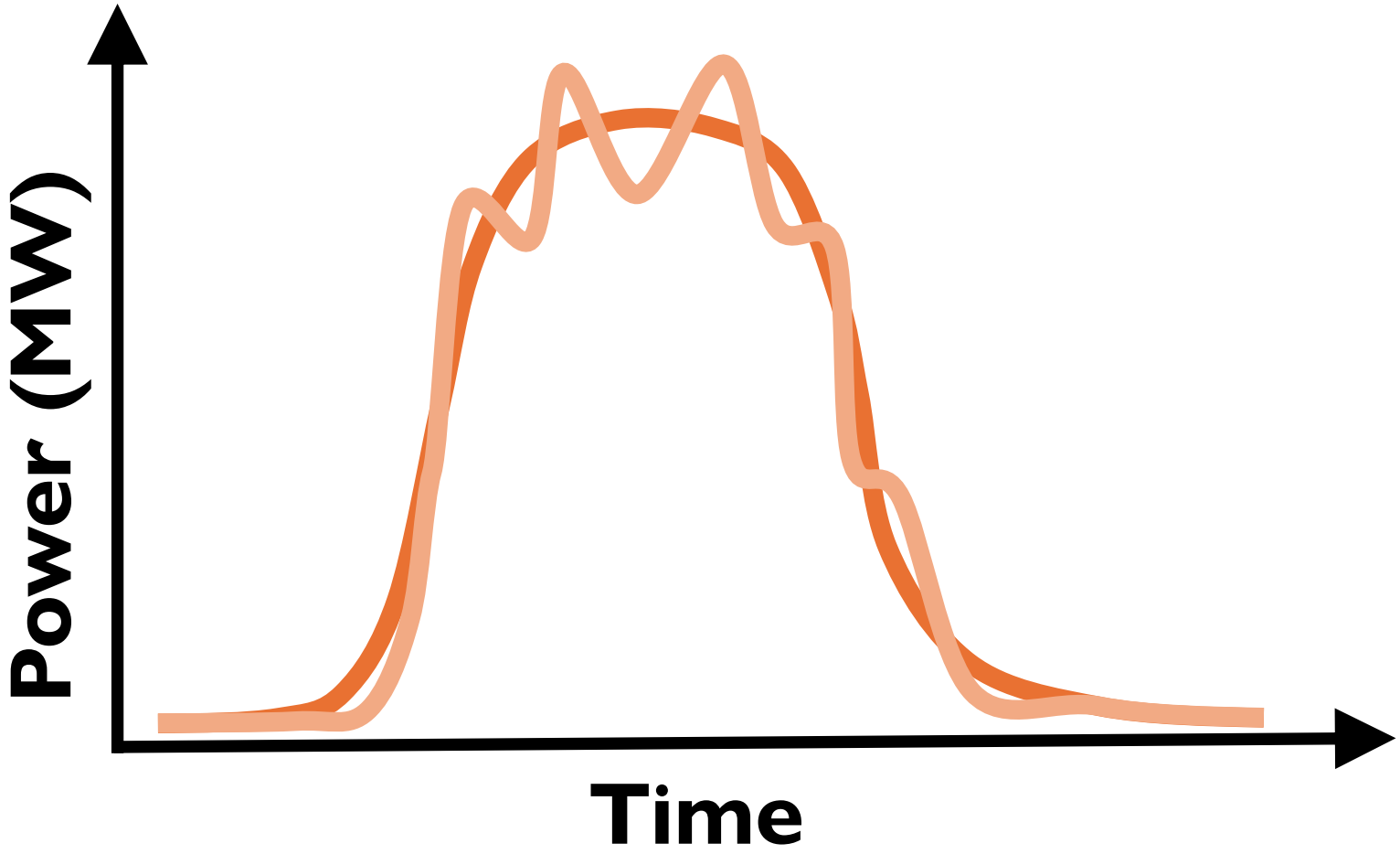
Operational variability



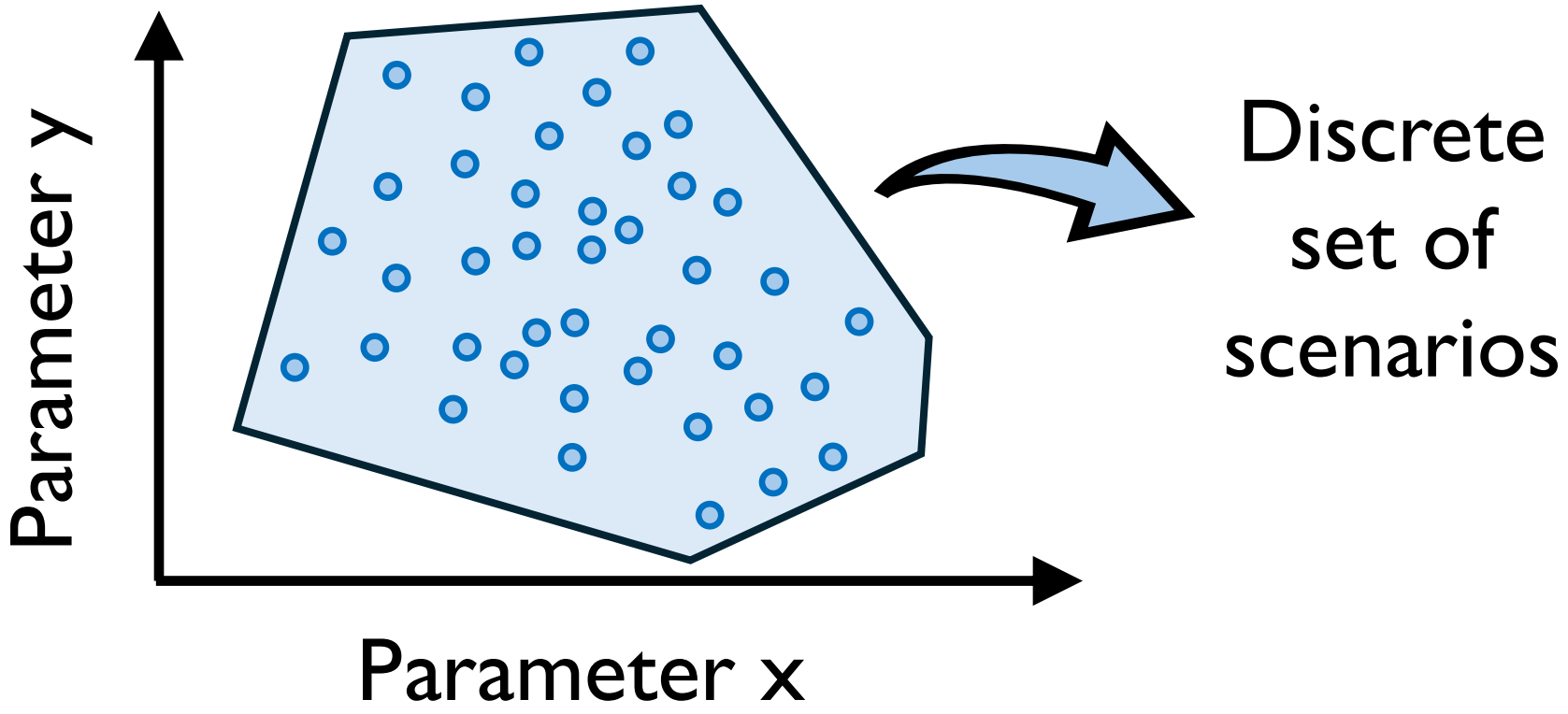


Uncertainty

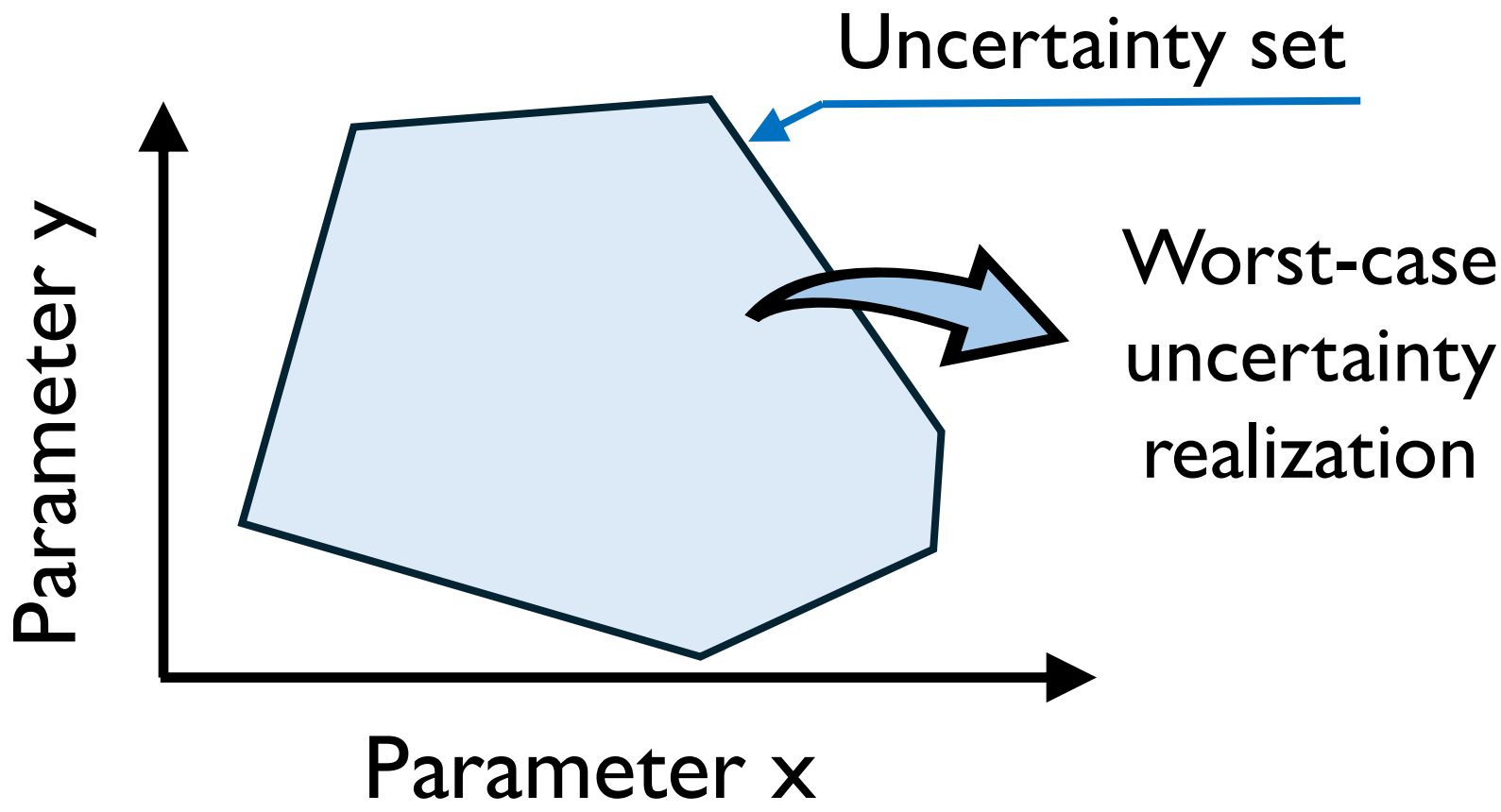




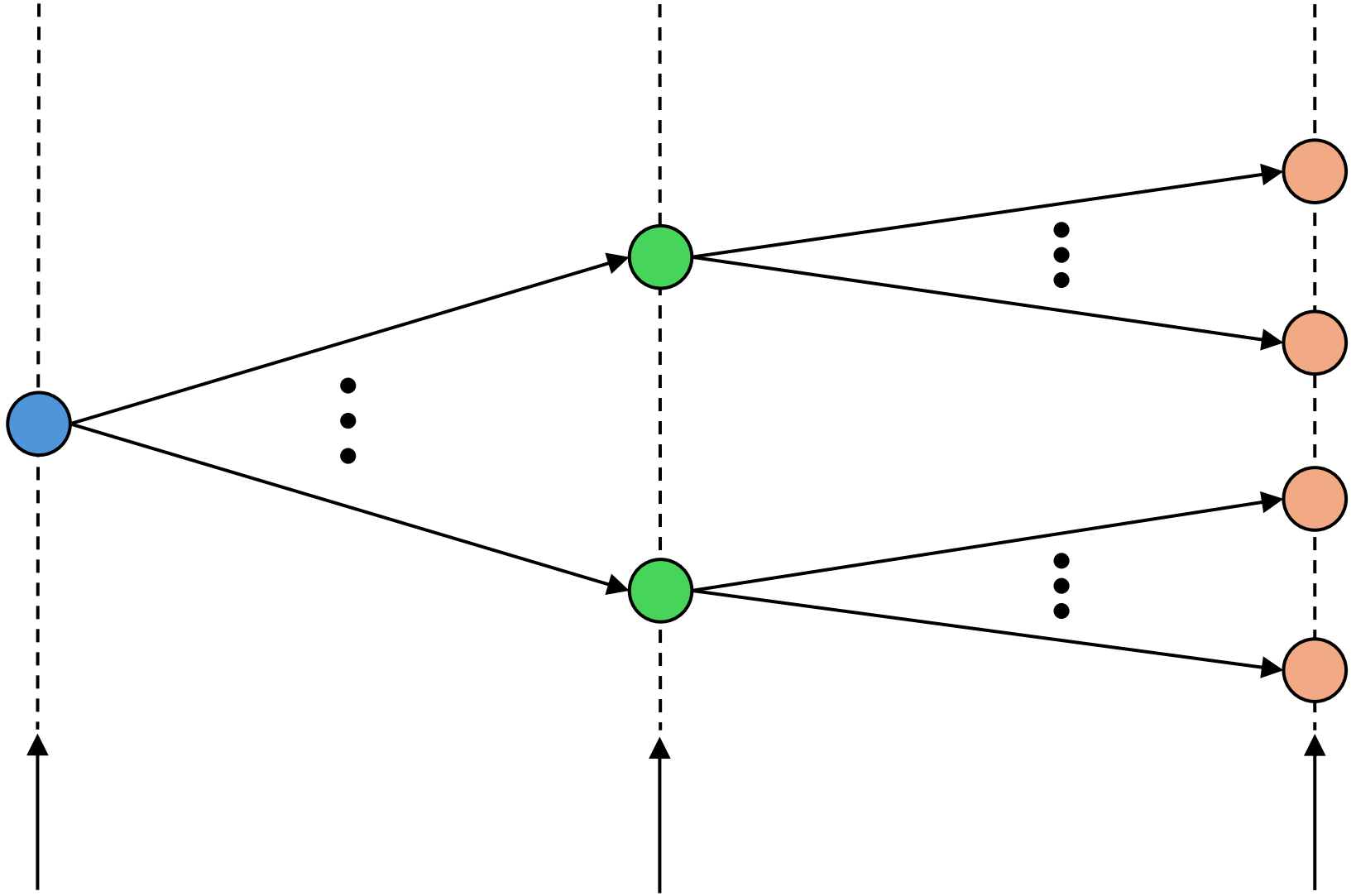
Stochastic programming



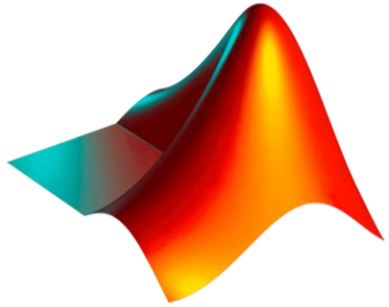
Robust optimization



Uncertainty



What is this course about?



MATLAB®

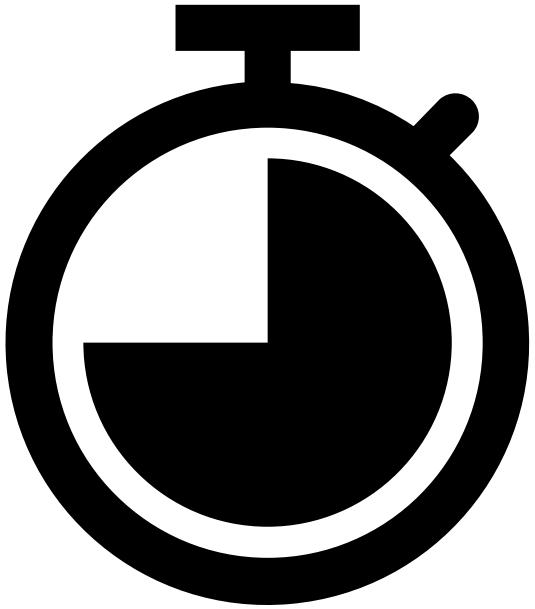


GAMS

What is this course about?

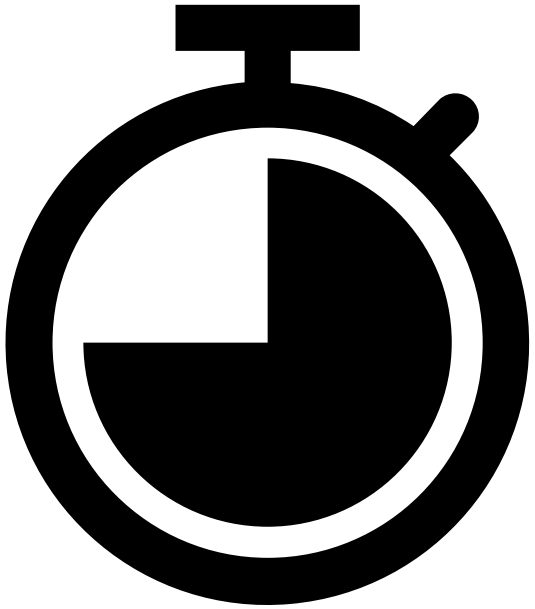
- Lecture 1: Introduction to Operations Research.
- Lecture 2: Operation and Planning Problems.
- Lecture 3: Operational Variability.
- Lecture 4: Uncertainty and Risk.
- Lecture 5: Further Applications.

What is this course about?



What would I explain to me if I were a doctoral student?

What is this course about?



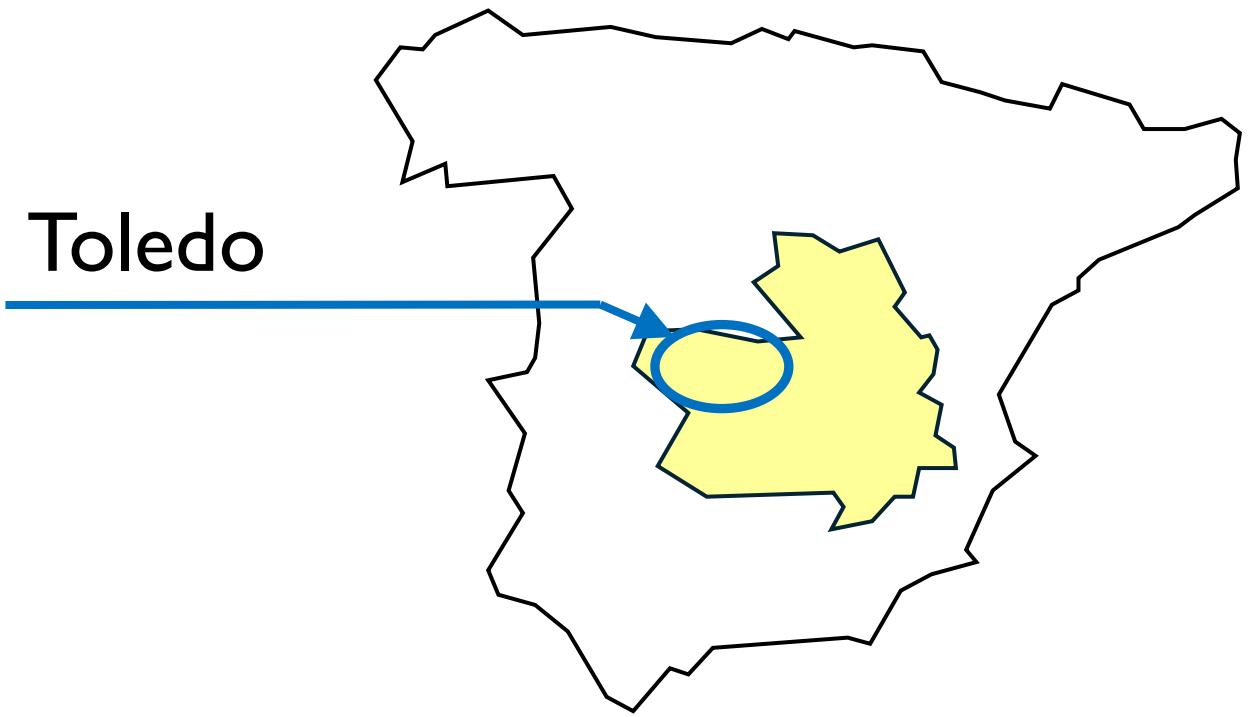
What would I explain to me if I were a doctoral student?

Develop new and better optimization tools

Who am I?

- Bachelor's Degree in **Electrical Engineering**.
- Master's Degree in **Industrial Engineering**.
- Ph.D. in **Industrial Engineering**.
- Postdoctoral Researcher.
- Assistant Professor.

Who am I?



UCLM
Universidad
Castilla-La Mancha

School of Industrial and Aerospace Engineering (Toledo)

Who am I?



Power and Energy
Analysis and Research
Laboratory

https://pearl.uclm.es/

Material of the course

https://github.com/AlvaroGarciaCerezo/CourseOR

